MILITARY SPECIFICATION

WASHER, KEY, METRIC
GENERAL SPECIFICATION FOR,

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flat long, flat long and short, and external types of washer, key, metric.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-613 - Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet and Strip)
QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Coating, Electrodeposited, Requirements For
PPP-H-1581 - Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of

MILITARY

MIL-F-495 - Finish, Chemical Black, For Copper Alloys
MIL-I-6868 - Inspection Process, Magnetic Particle
DOD-W-63546/1 - Washer, Key, Flat, Long, Steel, Metric
DOD-W-63546/2 - Washer, Key, Flat, Long, Brass, Metric

Beneficial comments (recommendations, additions, deletions) and any pertinent date which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover NJ 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document by letter.

FSC 5310

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STANDARDS

FEDERAL

FED-STD-66 - Steel: Chemical Composition and Hardenability

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1312 - Fasteners, Test Methods

(Copies of specifications, standards, handbooks, drawings, and publications required by maufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/
AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARD

ANSI/ASTM A 153 - Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A 109 - Steel, Carbon, Cold-Rolled Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19105.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Material. Recycled and reclaimed materials (see 3.1) shall be used to the maximum extent practicable.

3.2.1 Brass. Brass washers shall conform to the requirements of QQ-B-613, copper alloy no.268, temper one-half hard.

3.2.2 Steel. Carbon steel washers shall be of composition 1008 to 1018, cold rolled, no.4 temper, no. 3 finish in accordance with ASTM A 109.

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3.3 Hardness.

3.3.1 Carbon steel washers. Carbon steel washers shall have a maximum hardness of 65HRB. Washers having a thickness less than .75 mm shall have a maximum hardness of 60HR30T.

3.3.2 Brass washers. Brass washers shall have a maximum hardness of 70HRB. Washers having a thickness less than .75 mm shall have a maximum hardness of 65HR30T.

3.4 Finish. The protective finish of the washers when specified in the applicable specification sheet shall be as follows (see 4.5.3).

3.4.1 Black chemical. Brass washers shall be black oxide coated in accordance with MIL-F-495.

3.4.2 Cadmium plate. Carbon steel washers shall be cadmium plated in accordance with QQ-P-416, type II, class 3. The coating thickness shall be 5.1 um (micrometers).

3.4.3 Zinc coating. Carbon steel washers shall be zinc coated in accordance with QQ-Z-325 type II, class 3. The coating thickness shall be 5.1 um (micrometers).

3.4.4 Galvanized zinc. (hot-dipped). Carbon steel washers shall be hot-dipped galvanized in accordance with ASTM A 153.

3.5 Dimensions. Dimensions and tolerances shall be in accordance with the applicable specification sheet and as specified herein.

3.5.1 Parallelism. Washer faces shall be parallel within:

25 um for thickness less than 0.50 mm
50 um for thickness 0.50 mm to 1.00 mm
100 um for thickness 1.00 mm to 2.50 mm
150 um for thickness 2.50 mm to 4.00 mm

3.5.2 Flatness. Washer faces shall be flat within:

175 um for hole sizes thru 13 mm
250 um for hole sizes 14 mm thru 34 mm
380 um for hole sizes over 34 mm

3.5.3 Concentricity. The outside diameter (OD) of the washer shall be concentric with the inside diameter (ID) within the following limits as specified in table I.

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TABLE I. Concentricity.

Outside diameter (OD) mm		Concentricity
over	to	µm
1	3	125
3	6	150
6	10	180
10	18	215
18	30	260
30	50	310
50	80	370
80	120	435

3.6 Cracks. Washers shall be free from cracks, flaws, and pits in any location. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

3.7 Workmanship. Washers shall be free from surface contamination tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Material inspections (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 Material inspection. Material inspection shall consist of certification supported verifying data that the materials used in fabricating washers are in accordance with the applicable paragraphs listed in 3.2.1, 3.2.2, 3.3.1 and 3.3.2.

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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in table II.

TABLE II. Quality conformance inspections.

Inspection	Requirement paragraph	Test method paragraph
Dimensions	3.5	4.5.1
Magnetic particle (cracks)	3.6	4.5.2
Protective finish	3.4	4.5.3
Hardness	3.3.1, 3.3.2	4.5.4

4.4.1 Inspection lot. An inspection lot shall consist of completed washers covered by a single specification sheet, produced under essentially the same conditions and offered for inspection at one time.

4.4.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lost shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.4.3 Sampling for visual and dimensonal examination. Sampling shall be in accordance with inspection level S-3 of MIL-STD-105 with an acceptable quality level (AQL) of 2.5 percent defective for major defects. Minor defects shall be in accordance with inspection level S-3 of MIL-STD-105 with an AQL of 4.0 percent defective.

4.4.4 Sampling for finish test. Sampling for protective finish shall be in accordance with the applicable specifications referenced in 3.4.1 thru 3.4.4.

4.4.5 Sampling for other tests. Sampling for magnetic particle (cracks), and hardness inspection shall be in accordance with inspection level S-1 of MIL-STD-105 with an AQL of 1.5 percent defective.

4.4.6 Inspection of packaging. The sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with PPP-H-1581.

4.5 Methods of inspection.

4.5.1 Visual and dimensional examination. Samples taken as specified in 4.4.3 shall be thoroughly examined to determine conformance with this specification and applicable specification sheet. Examination shall be conducted in accordance with table III.

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TABLE III. Classification of defects.

Category	Defect	Inspection method
Critical	None defined	
Major		
101	Inside diameter, not as specified (see 3.5)	SIE 1/
102	Outside diameter, not as specified (see 3.5)	SIE
103	Thickness, not as specified (see 3.5)	SIE
104	Parallelism, not as specified (see 3.5.1)	SIE
105	Flatness, not as specified (see 3.5.2)	SIE
106	Evidence of cracks, flaws or pits (see 3.6)	Visual
Minor		
201	Protective finish, missing or incomplete (see 3.4)	Visual
202	Other dimensions, not as specified (see 3.5)	SIE
203	Workmanship (see 3.7)	Visual

1/ Standard inspection equipment.

4.5.2 Magnetic particle (cracks) inspection. When specified (see 6.2), samples taken as specified in 4.4.5 shall be subjected to magnetic particle inspection in accordance with MIL-I-6868. There shall be no evidence of cracks or pits as specified in 3.6.

4.5.3 Finish test. Carbon steel washers shall conform to the finish test of QQ-P-416, QQ-Z-325 or ASTM A 153 as specified in 3.4. Brass washers shall conform to the finish test of MIL-F-495.

4.5.4 Hardness test. Washers shall be tested for surface hardness in accordance with MIL-STD-1312, test method 6.

4.5.5 Packaging inspection. Sampling and inspection of the preservation-packaging, packing and container marking shall be in accordance with PPP-H-1581.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

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6. NOTES

6.1 Intended use. This specification covers various types of washers, key, flat, for general purpose, aircraft and general metric bolt and nut application. Their purpose is to provide an increase bearing surface and to prevent galling of the base material as well as safetying metric fasteners.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet
b. Applicable specification sheet part number
c. Magnetic particle (see 4.5.2)
d. Applicable levels of packaging (see 5.1)

Custodians:
Army - AR
Air Force - 11

Review activities;
Army - AV, AT, MI, EA
DLA - IS

User activities:
Army - ME
Navy - SH

Agent:
DLA - IS

Preparing activity:
Army - AR

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